

TIMELINE OF AI EVOLUTION

1950

ALAN TURING PROPOSES
THE TURING TEST



Alan Turing introduced the idea of testing whether a machine could imitate human intelligence through conversation. This is one of the early foundations of Artificial Intelligence.

1951

SNARC : Early Neural Network Machine



Marvin Minsky and Dean Edmonds built SNARC, one of the earliest neural-network machines. It showed the earliest onset of brain-inspired computing.

1956

Dartmouth Conference:
Birth of AI

The Dartmouth Conference is considered the official birth of artificial intelligence as a field. This was the event which started the AI as a formal research area.

1957 - 1959

General Problem
Solver

Allen Newell, Herbert Simon, and J.C Shaw developed the General Problem Solver. It was one of the first AI Programs designed to imitate human-like problem-solving.

MYCIN was developed at Stanford to help diagnose bacterial infections and recommend antibiotics. It showed that the AI could support expert decision-making in medicine.

MYCIN Expert System
1972 - 1976

Marvin Minsky and Seymour Papert published Perceptrons, which highlighted limitations of early neural networks. The reduced enthusiasm for neural network research helped lead towards the first AI winter.



Criticism of Early
Neural Networks

1969

Joseph Weizenbaum created ELIZA, one of the first chatbot programs. It stimulated conversation and became the early onset of natural language processing.



ELIZA : Early Chatbot

1966

Frank Rosenblatt developed the Perceptron, an early artificial neural network that could learn simple patterns. It became an important foundation for later neural networks and deep learning.



Frank Rosenblatt's
Perceptron

1958

1974 - 1980

FIRST AI WINTER

AI funding and interest declined because early systems failed to meet high expectations. Limited computing power, lack of data, and unrealistic predictions slowed progress.

1980s

Expert Systems
become commercial

Expert Systems became popular in business, medicine, finance, and engineering. They proved AI could have commercial value, but they were expensive and difficult to maintain.

1986

Backpropagation Revives
Neural Networks

David Rumelhart, Geoffrey Hinton, and Ronald Williams popularized backpropagation. This helped multi-layer neural networks learn more effectively and later became essential to deep learning.

1987 - 1993

SECOND AI WINTER

AI entered another downturn after expert systems became costly and hard to maintain. The collapse of the Lisp machine market also reduced to business confidence in AI.

ImageNet provided a massive labeled training and testing AI [models.it](#) became one of the key resources behind later computer vision breakthroughs.

Deep Learning gained momentum as better algorithms, larger datasets, and stronger computing power became available. This helped restart major progress in neural-network research

iRobot introduced Roomba, an autonomous robotic vacuum cleaner. It helped bring AI-powered robotics into everyday household life.

IBM's Deep Blue defeated world chess champion Garry Kasparov in a six-game match. It showed that AI could outperform humans in computer strategy games.



ImageNet Accelerates
Computer Vision

2009

Deep Learning Revival
Begins

2006



Roomba brings AI
Robotics Home

2002



IBM Deep Blue
Defeats Garry
Kasparov

1997

2011

IBM Watson Wins Jeopardy



IBM Watson defeated a former Jeopardy champions Ken Jennings and Brad Rutter. This was a major achievement in natural language processing and questions answering.

2011

SIRI Makes AI Voice Assistants Mainstream



Apple introduced Siri on the iPhone, bringing voice-based AI assistance to millions of users. This helped make AI part of everyday consumer technology.

2012

AlexNet wins ImageNet

AlexNet used deep convolutional neural networks and GPUs to achieve a major improvement in image recognition. This breakthrough helped launch the modern deep learning boom.

2014

GANs Advance Generative AI

Generative Adversarial Networks allowed AI systems to generate realistic images and synthetic media. This became an important step toward modern generative AI.

OpenAI released GPT-3, showing that very large language models could generate human-like text and perform many language tasks. It became a major step towards modern generative AI.

GPT-3 shows the Power of Large Language Models

2020

Google introduced BERT, a transformer-based model for natural language understanding. It improved search, question answering, and many other language AI tasks.

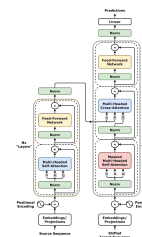
BERT Improves Language Understanding

2018

Google researchers introduced the Transformer architecture in Attention is All You Need. Transformers became the foundation for modern large language models.

Transformer Architecture Introduced

2017



Google Deepmind's AlphaGo defeated world Go champion Lee Sedol. This was important because Go is highly complex and required advanced deep thinking and reinforcement learning.

AlphaGo Defeats Lee Sedol

2016

2020

RAG connects AI with
External Knowledge

Retrieval-Augmented Generation, or RAG, combined language models with external information retrieval. This made AI more useful for search, enterprise knowledge systems, and factual questions answering.

2021

AlphaFold Transforms
Biology

DeepMind's AlphaFold made major progress in predicting protein structures. This showed how AI could accelerate biology, medicine, and drug discovery.

2022

Text-to-Image AI
becomes Mainstream

Tools such as DALL-E 2 and Stable Diffusion showed that AI could generate images from text prompts. This changed digital art, design, media and copyright discussions.

2022

ChatGPT brings Generative AI
to the Public

OpenAI released ChatGPT, making generative AI widely accessible through a conversational [interface.it](#) changed how people use AI in education, business, writing, coding, and research.

The EU AI Act showed that governments were beginning to regulate the AI. MCP showed the growing need for AI systems to connect safely with tools, files, databases and business systems.

EU AI Act and MCP Shape the Future
of AI

Claude showed the rise of safety-focused AI assistants, while LLaMA expanded the Open Source language [models.AI](#) agents also became popular as systems that could use tools and attempt multi-step tasks.

Claude, LLaMA, and AI Agents expand
the AI ecosystem

2024

2023